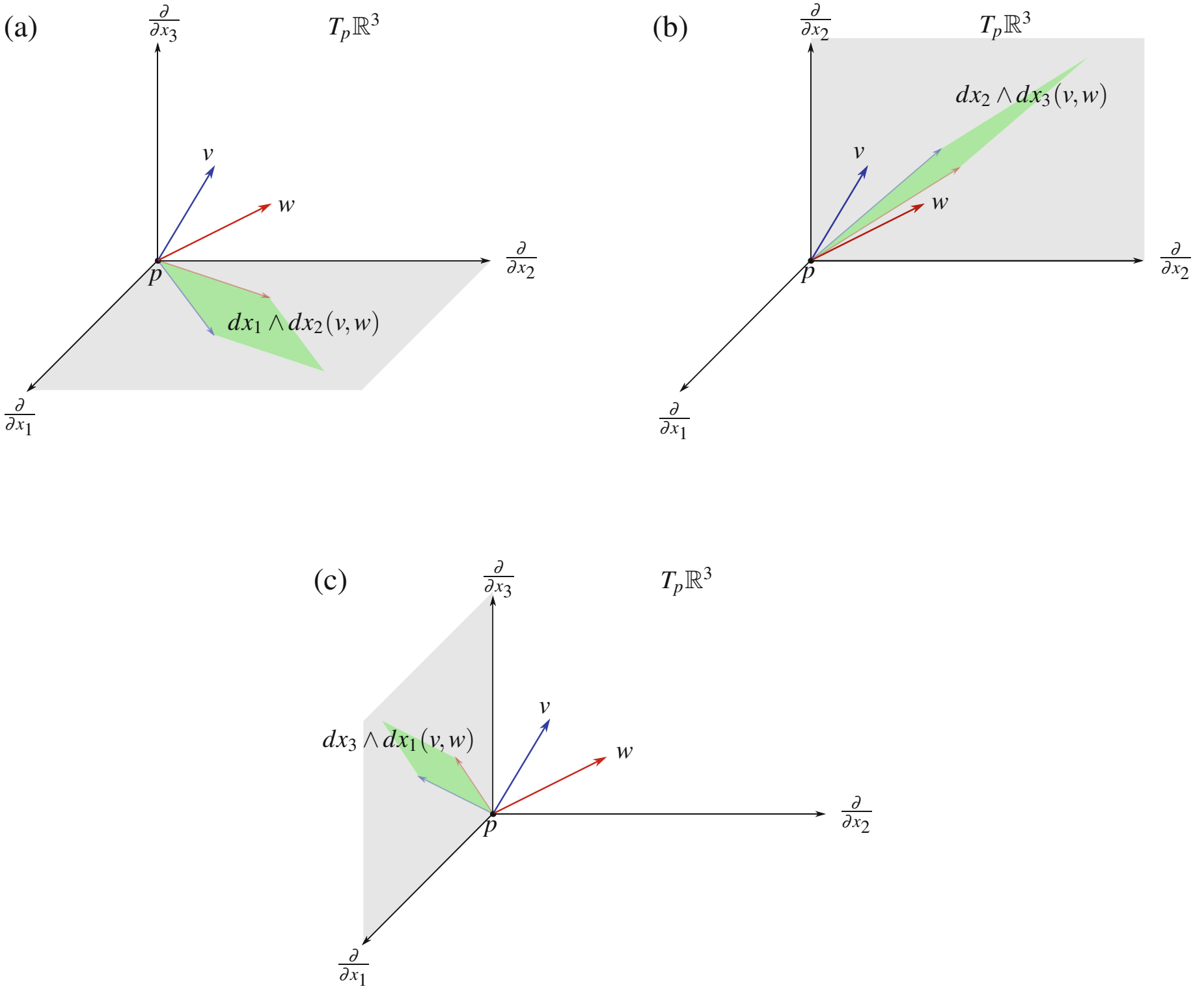

(a)
$\frac{\partial}{\partial x_3}$
$T_p\mathbb{R}^3$
v
w
$\frac{\partial}{\partial x_2}$
p
$dx_1 \wedge dx_2(v,w)$
$\frac{\partial}{\partial x_1}$
(b)
$\frac{\partial}{\partial x_2}$
$T_p\mathbb{R}^3$
$dx_2 \wedge dx_3(v,w)$
v
w
p
$\frac{\partial}{\partial x_2}$
$\frac{\partial}{\partial x_1}$
(c)
$\frac{\partial}{\partial x_3}$
$T_p\mathbb{R}^3$
$dx_3 \wedge dx_1(v,w)$
v
w
$\frac{\partial}{\partial x_2}$
p
$\frac{\partial}{\partial x_1}$